

Sensitive LPG Leakage Alarm

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Here is an ultra-sensitive LPG sensor that generates loud beeps when it senses any gas leakage. It detects vapours of liquefied petroleum gas anywhere between 200 and 10,000 ppm and drives a piezobuzzer to catch attention for immediate action. The buzzer beeps until the concentration of gas in the air decreases to a safe level. The circuit uses an MQ6 gas sensor, which is designed to sense LPG, propane and isobutane gases.

Circuit and working

Fig. 1 shows the circuit of the LPG sensor. The circuit is built around 5V voltage regulator 7805 (IC1), gas sensor MQ6 (GS1), counter IC 4060 (IC2) and

a few discrete components.

GS1 is a six-pin gas sensor that can detect very small traces of LPG in the air and has a swift response time. However, it has very less sensitivity to alcohol and smoke. The sensor's output is in the form of resistance.

As indicated in Fig. 1, the pins of GS1 are H, A and B, two each on either side. H pins are for the heater with no

Test Points

Test point	Details
TP0	0V, GND
TP1	12V
TP2	5V
TP3	0V when sensor is exposed to LPG and LED1 blinks
TP4	5V square pulse

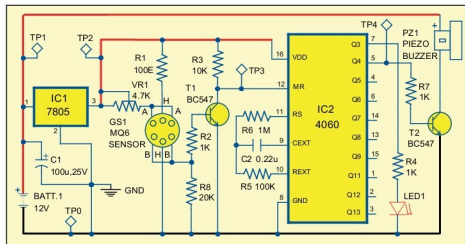


Fig. 1: Circuit of the sensitive LPG sensor

polarity. Input pins A or B and output pins A or B can be connected either way round.

The coil heater inside the sensor can be easily heated with 5V DC. If pin A is connected to 5V DC through variable resistor VR1, use pin B as the output or vice versa. Both A and B pins can be shorted. In short, H pins are connected to positive and negative rails, A or B pin to 5V DC, and B or A for output.

The resistance value of GS1 is different for various kinds and concentration of gases. So when using this sensor, sensitivity arrangement is very important. For accurate detection, it is necessary to calibrate the sensor for 1000 ppm of LPG concentration in the air with load resistance of about 20 kilo-ohms. (In the datasheet, the load resistance range of MQ6 is mentioned as 10 kilo-ohms to 47 kilo-ohms.)

Preset VR1 is used to adjust the sensitivity of the sensor to a particular gas concentration. Output from the sensor is connected to the base of transistor T1, which acts as a switch to trigger the alarm generator built around IC2.

IC2 is a binary counter IC that oscillates using capacitor C2 and resistor R5. Transistor T1 controls the reset pin (pin 12) of IC2. When



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